

Highlights

Large per-window oracle headroom that three selector families did not convert: a feedback-delay audit of plug-in selection for time series forecasting

Mianhan Liu, Chen Chen

- Block filtering, metric and mean-vs-median each flip a plug-in verdict.
- Per-window oracle headroom survives per-window arm relabeling.
- Best-arm identity has a median half-life of 5.1% of the forecast horizon.
- A quarter-wise lead-time oracle is worth only 0.24% over a frozen arm.
- Dataset-clustered intervals and leave-one-dataset-out bound every verdict.

Large per-window oracle headroom that three selector families did not convert: a feedback-delay audit of plug-in selection for time series forecasting

Mianhan Liu^{a,*}, Chen Chen^a

^aIndependent Researcher, Shanghai, 200000, China

ARTICLE INFO

Keywords:

time series forecasting
model selection
delayed feedback
benchmarking
reproducibility
negative result

ABSTRACT

Model-agnostic “plug-and-play” modules for time series forecasting—instance normalization, non-stationarity correction, frequency-domain losses—are reported as helpful, yet their gain flips sign across backbones. The natural response is to learn *when* to attach *which* module. In 2,035 single-GPU training runs across four backbones, eight datasets and four horizons, three selector families did not convert that promise into a deployable gain. First, the effect is smaller than three routine reporting choices—complete-block filtering, the error metric, and mean versus median—each of which flips a verdict. Second, the per-window oracle motivating gating is provably invariant to re-labeling which arm produced which error, so its +10.15% headroom is no evidence of learnability, although 79% of it survives a seed change. Third, best-arm identity has a median correlation half-life of 5.1% of the horizon, whereas deployment forces a decision lag of at least one horizon. The selectors lose to the arm frozen on validation, by 0.87 and 0.67 points, and are absolutely worse in most groups. With the eight datasets as the unit of generalization, the gating loss survives clustering while the online loss no longer excludes zero. Only a constant mis-selection was recoverable; we distill the audit into a checklist.

1. Introduction

The past five years have produced a steady stream of *model-agnostic* additions to deep time series forecasters: reversible instance normalization [1], explicit non-stationarity correction [2], channel clustering [3], frequency-domain training objectives [4]. Each is presented as an orthogonal, architecture-independent, essentially free improvement. Two facts sit uneasily beside that framing.

First, benchmarks have saturated. On ETT / Electricity / Weather, the annual improvement of the best published model has fallen from double digits to a few percent, and unified re-evaluations attribute much of the remaining spread to pipeline details rather than to the method itself [5, 6]; in that regime a reported +0.3% is indistinguishable from a reporting choice.

Second, the sign of the gain is unstable. Practitioners who attach the same plug-in to a linear model and to a patch transformer routinely observe gains of opposite sign. If the gain is conditional, the useful engineering artifact is not another plug-in but a *decision rule*: given (backbone, dataset, horizon), or even given the current window, decide whether to attach a plug-in and which one.

This paper is a systematic, negative, and — we argue — *mechanistic* answer to that decision problem, within a protocol we state explicitly. We do not propose a new plug-in. We build the full conditional-gain matrix, quantify how much of the reported effect is produced by reporting conventions rather than by the method, attack the decision problem with three independent selector families, and offer a quantitative explanation for their uniform failure: **on this grid the correlation length of best-arm identity is one to two orders of magnitude shorter than the feedback delay that deployment imposes.**

Our contributions are fourfold.

*Corresponding author.

✉ 1367489893@qq.com (M. Liu); 1577027526@qq.com (C. Chen)

ORCID(s): 0009-0000-0274-1062 (M. Liu); 0009-0001-1991-9370 (C. Chen)

1. **Three reporting levers, each larger than the effect.** Complete-block filtering of the main table, the choice of error metric, and the choice between a mean and a median each move the headline by more than the effect being claimed, and each flips at least one plug-in verdict (§4). A fourth, reporting a plug-in only at its published default hyper-parameter, accounts for essentially all of one plug-in’s apparent benefit (§5).
2. **Effect size in interpretable units.** We put the block-level plug-in effect on the same scale as the run-to-run noise of a training seed and as the compute it costs (§6): 28% of block-level effects are smaller than same-configuration seed noise, and the arm with the largest compute overhead is also the only one that is significantly *worse* than doing nothing. The seed floor is measured on the 208 of 428 configurations we could afford to replicate, all on four cheap datasets, and we report that restriction wherever we use it.
3. **A label-invariance argument, and the quantity it points at.** We prove and empirically verify that the per-window oracle used to motivate gating is invariant to per-window arm re-labeling (§8), so a large oracle number is not evidence that anything is learnable. We then measure the quantity that does govern learnability — the lagged predictability of best-arm identity — and find it at chance well before the legal decision lag in every group we measured (§8.2). A seed-paired replication rules out the trivial explanation that the headroom is training noise (§8.3).
4. **A falsified-hypotheses table.** Three selector families and two control conditions, all evaluated on identical decision groups against a deployable baseline, are all negative, with a delay sweep locating the break-even point at $0.25 \times H$ (§9). We also audit the one selection axis that is *not* subject to feedback delay, namely the lead-time index within a forecast (§7); a quarter-wise oracle on that axis is worth only +0.24%. Neither axis produced a deployable gain here, for two different reasons.
5. **Dataset-level inference.** A claim about “plug-in selection” is a claim that generalizes across datasets, so the eight datasets, not the 123 decision groups, are the units it rests on. We therefore report every headline effect three ways — pooled over groups, as a dataset-cluster bootstrap, and under leave-one-dataset-out — and we say which conclusions narrow (§10). The oracle headroom and the gating loss survive; the online loss does not exclude zero once the datasets are the resampling unit.

We deliberately report the deployable baseline `best_fixed_val` — the arm chosen on validation data and then frozen — rather than the post-hoc best arm. Much of the apparent success of adaptive selection in the literature disappears against this baseline, and §11 shows that what remains is a validation→test transfer effect, not adaptation.

What we do not claim The result is an audit of a protocol, not a theorem about selection. We do not show that per-window plug-in selection is impossible, that no selector can work, or that this direction is closed. What we show is bounded by four things, and every conclusion below should be read inside them. *Arms*: three plug-ins (`revin`, `san_lite`, `fredf`) plus the unmodified backbone, on four backbones, eight datasets and eight horizons. *Feedback model*: the selector observes a window’s realized error only after the full horizon has elapsed, which is the correct model for a forecaster whose target is a horizon of future values and the wrong model for a task where partial feedback arrives earlier. *Selectors*: three families — static complexity-feature gating, delayed online error feedback, horizon-aware checkpoint selection — each with the hyper-parameter sweeps reported in §9, and not, for instance, a meta-learned selector trained across datasets or one given exogenous covariates. *Statistics*: MSE and MAE on the standard splits, with decision groups as the primary aggregation unit and datasets as the unit for any cross-dataset claim. A selector outside these boundaries is not addressed by our evidence, and §11 lists the two openings we consider most promising.

2. Related work and positioning

Plug-in modules. Reversible instance normalization [1] removes instance-level distribution shift by de-normalizing the output with the input statistics. Non-stationarity correction methods [2] estimate slice-wise statistics and extrapolate them; we implement a *training-free surrogate* (`san_lite`, EWMA extrapolation of slice statistics) so that the arm adds zero trainable parameters and zero per-dataset hyper-parameters,

and we label it as a surrogate throughout rather than claiming to reproduce [2]. Frequency-domain losses [4] (**fredf**) add a spectral term with weight α . Our contribution to this literature is not a new member but an audit of how their gains are reported.

Benchmark-hygiene work. A line of work has shown that apparent progress in long-horizon forecasting is sensitive to pipeline details such as **drop_last**, look-back length and checkpoint selection, and that conclusions change once evaluation is unified [5, 7, 8]; a recent reproducible re-evaluation of eight models on fourteen datasets reports that small changes to the experimental setup or to the evaluation metric are enough to overturn claimed state-of-the-art results [6]. We re-run every baseline ourselves under one fixed protocol for exactly this reason and never copy numbers across papers. Section 4 adds new items to that list: which *blocks* enter the aggregate, which *metric* is reported, and which *central estimator* is used are each a significance lever, and we follow [9] on how block-level comparisons should be tested.

Adaptive / online model selection. Expert-advice and bandit selection over forecasters is classical [10], and dynamic expert-arbitration for time series is an established family [11], as is drift-triggered adaptation [12]. The regret guarantees, however, assume feedback arrives shortly after the decision; delayed-feedback online learning quantifies the degradation when it does not [13]. The novelty here is empirical and negative: in H -step-ahead forecasting the feedback delay is *structurally* H windows, and we measure the identity correlation length that this delay must be compared against. To our knowledge the ratio half-life/ H has not previously been measured for plug-in selection; it is the quantity that decides the problem.

3. Protocol

Grid. Backbones: **DLinear** [8], **PatchTST** [14], **TimesNet** [15], **iTransformer** [16] (imported unmodified from a standard library [7]; we do not use its training script). Datasets: **ETTh1**, **ETTh2**, **ETTm1**, **ETTm2**, **Electricity**, **Exchange**, **Weather** (horizons 96/192/336/720, look-back 96) and **ILI** (horizons 24/36/48/60, look-back 36). Arms: **none**, **revin** [1], **san_lite** (surrogate for [2]), **fredf** [4] ($\alpha=0.5$), plus an α -sweep pool $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ on the plain objective and $\alpha \in \{0.1, 0.5, 0.9\}$ on the $\sqrt{H}$ -normalized one (§5).

Three phases. Phase 1 is the main gain matrix: 844/844 cells. Phase 2 persists per-window squared errors on validation and test for the nominal four-arm pool: 423/428 cells (5 missing cells are all **TimesNet** $\times H=720$). Phase 3 is the α sweep over three backbones (**TimesNet** excluded): 768/768 cells, i.e. 96 blocks $\times$ 8 arms (seven α settings plus a re-trained **none** control). In total we perform 2,035 training runs on a single V100 GPU, in fp32 with AMP enabled on the largest datasets. The three phases overlap by design (2,035 counts runs, not distinct configurations: phase 1 includes 3 seeds on a subset, and phase 2 re-runs the four-arm pool to persist per-window traces).

Statistical block. A block is a (dataset $\times$ horizon $\times$ backbone) triple, not a dataset. This choice is not cosmetic. With $k=4$ methods and only the $N=8$ datasets as blocks — the aggregation level most plug-in papers use — the Nemenyi critical difference is 1.66, wider than the entire observed rank range, so no ranking can be significant at that resolution. At our $N=128$ blocks it falls to 0.41. We follow [9] for the Friedman/Nemenyi machinery, and all paired tests are Wilcoxon signed-rank reported both raw and with Holm correction inside an explicitly declared family.

A documented revin pruning. **PatchTST**, **TimesNet** and **iTransformer** hard-code their own instance normalization, so wrapping them in an external **RevIN** is mathematically an approximate identity. We therefore pruned that combination *a priori* (skip rule **revin_on_internal_norm_backbones** in **configs/matrix.yaml**) and kept a verification subset — **{ETTh1, Weather}** $\times$ **{96, 720}** on all three backbones, 12 cells — to check the prediction instead of assuming it. It holds: the median gain over **none** on those 12 cells is +0.00% (Wilcoxon $p=0.23$), and on **PatchTST** and **iTransformer** the test MSE is unchanged to four decimals in 7 of 8 cells; the only two cells with a visible effect are **TimesNet** at $H=720$ (+0.62% and +5.56%). On **DLinear**, which has no internal normalization, the same arm has a median gain of +0.98% over 64 cells. No cell of the matrix failed to train — all 844 phase-1 runs report status **ok** — so every **revin** absence below is this design decision, not an outage. The consequence for §4 is mechanical but must be stated: because **revin** exists on 44 blocks by construction, complete-block filtering over four arms can only ever return those 44 blocks.

Table 1

Same raw runs, two aggregation conventions. Mean relative gain over **none** (%), Wilcoxon signed-rank p raw and Holm-adjusted within the family of three arms (each convention is its own family). The significance flip is not an artifact of multiplicity: after Holm correction the complete-block convention still calls **fredf** significant (0.017) while the correct pairwise convention does not (0.76).

Convention	Arm	N	mean	p	p_{Holm}
complete-block (44)	fredf	44	+3.60	0.0058	0.017
pairwise	fredf	128	− 2.37	0.44	0.76
complete-block (44)	san_lite	44	−1.29	0.056	0.11
pairwise	san_lite	128	− 3.94	5×10^{-8}	1.5×10^{-7}
complete-block (44)	revin	44	+3.55	0.38	0.38
pairwise	revin	44	+3.55	0.38	0.76

Decision group. For window-level analysis a *decision group* is a (backbone, dataset, horizon) triple for which all arms have aligned per-window error vectors on the same evaluation windows. We have 127 groups for test-side audits. For protocols that must *train* a gate on validation windows we require a provenance marker certifying deterministic loader order (§13); 96 groups qualify. The nominal pool has four arms, but under the pruning above the *realized* arm count per group is 4 in 43 groups, 3 in 83 and 2 in 1 (for the 96 gating groups: 40 and 56). Since oracle headroom grows with the number of arms, we report the realized counts wherever a headroom figure appears and never pool groups of different sizes into a single headroom claim without saying so.

Baselines. **none**; **best_fixed_val**, the arm with lowest validation MSE, frozen over the test segment (this is the deployable strong baseline and the reference for every gain we report); **best_fixed_test**, the post-hoc constant arm (an upper bound on any non-time-varying policy); and the per-window **oracle**.

4. How the gain is reported: three significance levers

4.1. Lever 1: which blocks enter the aggregate

The standard way to aggregate a plug-in comparison is to keep the blocks in which *all* arms are available and run a paired test on that subset. Absent cells, however, are never absent at random. Whatever prunes a cell — an out-of-budget corner (heavy backbone $\times$ long horizon $\times$ large dataset), a crash, or, as here, an explicit design rule — is correlated with the very axes along which the plug-in’s effect varies. In our matrix, 128 blocks exist, but only 44 are complete across four arms, because the **revin** pruning of §3 decides the subset; 32 of those 44 (73%) come from **DLinear**, the one backbone without internal normalization, which is also the backbone on which frequency-domain losses happen to help most. The filter is therefore a backbone filter wearing a completeness costume, and the arm it silently re-weights is not the arm that caused the pruning.

Table 1 contrasts the two conventions. Pairwise dropping (each arm compared to **none** on every block where *that pair* exists) is the correct estimator; complete-block filtering is what the literature does.

The direction of the error is not benign. For **fredf** the filter converts a non-significant negative mean into a significant positive one; for **san_lite** it hides a real, strongly significant degradation. On the complete-block subset a Friedman test rejects ($\chi^2=30.9$, $p=8.9 \times 10^{-7}$, $CD=0.71$) with mean ranks **fredf** 1.93 < **revin** 2.14 < **none** 2.59 < **san_lite** 3.34 — a clean-looking ranking produced by a subset that is 73% one backbone.

Table 2 shows why the aggregate is so fragile: conditioned on the backbone, the same plug-in ranges from +4.6% to −16.6%. *Plug-in gains are strongly backbone-conditional*, which is the empirical premise that makes a selection rule attractive in the first place. The rest of the paper tests whether that premise can be converted into a deployable decision.

Table 2

fredf vs. **none** per backbone, all pairable blocks. The sign flips and the magnitude spans 21 points. p_{Holm} is adjusted within the family of four strata; the two significant cells have opposite signs.

Backbone	N	mean (%)	p	p_{Holm}
DLinear	32	+4.56	0.0036	0.011
PatchTST	32	-16.63	3.3×10^{-9}	1.3×10^{-8}
TimesNet	32	+1.41	0.15	0.31
iTransformer	32	+1.17	0.26	0.31

Table 3

Metric robustness of the plug-in verdicts.

Arm	Metric	N	W/T/L	Median gain (%)	95% CI	p	p_{Holm}	r	Verdict
revin	MSE	44	23/0/21	+0.03	[-0.52, +0.84]	0.382	0.764	0.13	n.s. (better)
	MAE	44	37/0/7	+1.38	[+0.43, +4.05]	4.8×10^{-6}	1.4×10^{-5}	0.65	sig. (better)
cross-metric: Spearman $\rho = 0.64$, sign agreement 64%, verdict flips									
san_lite	MSE	128	30/0/98	-3.01	[-4.35, -2.05]	4.9×10^{-8}	1.5×10^{-7}	0.48	sig. (worse)
	MAE	128	44/0/84	-1.01	[-1.60, -0.65]	8.7×10^{-4}	0.002	0.29	sig. (worse)
cross-metric: Spearman $\rho = 0.84$, sign agreement 84%, verdict consistent									
fredf	MSE	128	70/0/58	+0.45	[-0.45, +1.27]	0.438	0.764	0.07	n.s. (better)
	MAE	128	79/0/49	+0.90	[+0.41, +1.57]	0.323	0.323	0.09	n.s. (better)
cross-metric: Spearman $\rho = 0.97$, sign agreement 93%, verdict consistent									

Paired Wilcoxon signed-rank per arm against **none**; blocks are (dataset $\times$ horizon $\times$ backbone), matched pairwise so that one arm's pruning cannot shrink another arm's sample. Holm correction is applied within each metric (family size 3). CIs are percentile bootstrap over blocks (10^4 resamples, blocks resampled jointly to preserve pairing). Positive gain = lower error than **none**.

4.2. Lever 2: which error metric is reported

Every number above is MSE, as is standard. Because our evaluation accumulates MAE in the same forward pass, re-running the entire test on MAE costs nothing, and the outcome is not a formality (Table 3). The verdict for **revin** *flips*: it is indistinguishable from **none** on MSE (median +0.03%, $p_{\text{Holm}}=0.76$) and significantly better on MAE (median +1.38%, $p_{\text{Holm}}=1.4 \times 10^{-5}$, winning 37 of 44 blocks). The flip is not a ranking artifact: the per-block gains on the two metrics correlate at Spearman $\rho=0.64$ but agree in sign in only 64% of blocks, i.e. **revin** shifts the error distribution in a way that a squared loss and an absolute loss genuinely disagree about. This is the expected signature of a normalization module: it trims many moderate errors while occasionally enlarging a few large ones, which MAE rewards and MSE does not. For **fredf** and **san_lite** the two metrics agree ($\rho=0.97$ and 0.84).

4.3. Lever 3: mean or median, and the tail that hides behind it

A third choice is left implicit almost everywhere: papers report a *mean* relative gain and test it with a *rank* test. The two target different functionals, and on this data they disagree about whether a plug-in works at all (Table 4). For **fredf** the mean is -2.37% while the median is +0.45%, and the same conflict appears on MAE (-1.18% vs. +0.90%). Neither number is wrong; they answer different questions, and a paper that prints the mean while testing the median is not testing what it reports.

The mechanism is a heavy left tail (skew -4.5), and the tail is not diffuse: it is one identifiable corner of the grid. **fredf** degrades **Electricity** with PatchTST by -31%, -61%, -96% and -123% at $H=96, 192, 336, 720$, i.e. the failure grows monotonically with the horizon, while the same plug-in helps DLinear at $H=720$ by +19% on ETm2 and +25% on Exchange. Overall 10.9% of blocks are more than 10% worse than **none** and 3.9% are more than 25% worse. Reporting a median alone would sell **fredf** as a small free win; reporting a mean alone would sell it as harmful; reporting either alone hides a deployment risk that a practitioner needs to know about. **san_lite** carries the highest frequency of moderate damage (21.9% of blocks below -10%) but its worst case is milder (-30%).

Table 4

The choice of central estimator is a third significance lever.

Arm	Metric	Mean (%)	Median (%)	Mean—Med	Skew	< −10%	< −25%	Worst block	Worst (%)
revin	MSE	+3.55	+0.03	+3.52	+2.19	2.3%	0.0%	Exchange/720/DLinear	-14.1
	MAE	+5.01	+1.38	+3.63	+1.57	0.0%	0.0%	ETTh1/720/TimesNet	-5.9
san_lite	MSE	-3.94	-3.01	-0.93	+0.96	21.9%	2.3%	ILI/36/TimesNet	-30.0
	MAE	-1.08	-1.01	-0.07	+1.12	7.8%	0.0%	ILI/60/TimesNet	-19.4
fredf	MSE	-2.37	+0.45	-2.82	-4.47	10.9%	3.9%	Electricity/720/PatchTST	-123.4
	MAE	-1.18	+0.90	-2.07	-4.60	6.2%	3.1%	Electricity/720/PatchTST	-80.0

All columns are computed on the *same* pairwise-matched blocks as Table 3; only the summary functional changes. Bold rows are sign conflicts, where the mean and the median disagree on whether the plug-in helps at all. “< −10%” is the fraction of blocks whose error is more than 10% worse than none. A paired Wilcoxon test targets the median, so a paper that reports a mean and tests with Wilcoxon is not testing the quantity it reports.

Table 5

The apparent benefit of a frequency-domain loss is recovered by per-cell tuning of its one hyper-parameter; at the published default the arm is harmful. 96 blocks, three backbones (TimesNet excluded).

Setting	mean gain (%)
fixed $\alpha=0.5$ (published default)	−3.70
per-cell α tuned on <i>validation</i>	+1.79
per-cell α tuned on <i>test</i> (cheating bound)	+2.29
tuning gap	+5.48
best <i>single</i> α chosen on validation ($\alpha=0.1$)	+0.12
with $\sqrt{H}$ normalization ($\alpha=0.1$)	+0.18

We therefore report median, mean, a paired bootstrap interval and the tail fractions together for every arm, and we recommend the same in §12. Note that the three levers are independent: they act on which blocks are counted, which loss is measured, and how the counted blocks are summarized, so a paper can be internally consistent and still be off by more than its own effect size on all three at once.

5. How the gain is tuned: “plug-and-play” is a tuning effect

Frequency-domain losses introduce one hyper-parameter, the spectral weight α , and are reported at a fixed default. We sweep $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ on the plain objective and $\alpha \in \{0.1, 0.5, 0.9\}$ on the $\sqrt{H}$ -normalized one, giving $96 \times (5 + 3) = 768$ α -cells for analysis. Of these, the 96 plain $\alpha=0.5$ cells are the phase-1/2 reference and the remaining 672 are trained in phase 3; phase 3’s own run count is also 768 because it re-trains its own **none** control ($96 \times (7 \alpha \text{ settings} + 1)$), so the two 768s count different things and are not a duplicate. The 96 blocks are 3 backbones $\times$ 8 datasets $\times$ 4 horizons — TimesNet is excluded from the sweep for cost reasons, so this section does not claim to cover it.

Two readings follow from Table 5. (a) The method is not plug-and-play: essentially all of its *aggregate* benefit is recovered by per-cell tuning. Tuning raises the share of positive blocks from 46.9% at $\alpha=0.5$ to 59.4%, and 14.6% of blocks (14 of 96) flip from non-positive to positive; two blocks flip the other way, because the tuning is honest and uses validation data only. So the default is wrong for a large minority of cells rather than for all of them, and the aggregate sign change is driven by magnitudes as much as by counts. (b) The tuning is genuinely per-cell: no single α transfers (+0.12%), and the number of distinct optimal α values that one (backbone, dataset) pair takes across horizons is 1.79. Normalizing α by $\sqrt{H}$ reduces that to 1.42 and adds 0.06 points — we report it as an ablation line, not as a contribution.

The gap between validation-tuned (+1.79) and test-tuned (+2.29) is only 0.5 points, so honest tuning is nearly as good as cheating; the problem is not selection noise, it is that the default is wrong for most cells.

Table 6

Plug-in effects against seed noise and against their compute cost.

Arm	N_{noise}	Within seed noise	Median $ \Delta /\sigma_{\text{seed}}$	Train-time ratio	Median gain (%)
revin	16	19%	3.77	1.05×	+0.03
san_lite	64	27%	2.53	1.73×	-3.01
fredf	64	31%	1.74	1.03×	+0.45

Left half: for the 144 block-arm pairs where both the arm and none have ≥ 2 seeds, the noise scale is $\sigma_{\text{seed}} = \sqrt{s_{\text{arm}}^2 + s_{\text{none}}^2}$ (median $\sigma_{\text{seed}} = 0.0049$ MSE). “Within seed noise” is the fraction of blocks whose entire plug-in effect is smaller than that scale, i.e. a different seed could flip its sign. Overall 28% of block-level effects and 27% of the wins fall in this regime. Right half: training wall-clock ratio against none in the same block.

6. How big the gain is: seed noise and compute cost

Before asking whether a selector can exploit these gains, it is worth asking how large they are in units a practitioner can act on. Two cheap references make the effect size interpretable, and both are computable from artifacts we already have (Table 6).

Against seed noise. For 208 of the 428 phase-one configurations we trained three seeds, giving 144 block-arm pairs in which both the arm and none have a within-configuration standard deviation. This coverage is not uniform and the restriction matters for what follows: the three-seed configurations are exactly the four cheap datasets (ETTh1, ETTh2, Exchange, ILI), and ETTm1, ETTm2, Electricity and Weather contain no replicated configuration at all, so 58 of 128 blocks (45.3%) have all four arms replicated. We compare the plug-in effect against $\sigma_{\text{seed}} = \sqrt{s_{\text{arm}}^2 + s_{\text{none}}^2}$, the noise scale of a difference of two seed-averaged means; its median is 0.0099 MSE, while the median single-arm seed standard deviation — a smaller quantity that is easy to confuse with it — is 0.0049 MSE. In relative terms the within-configuration dispersion of MSE has median 0.95%, interquartile range 0.30–2.41% and 90th percentile 4.55%, which is the band that any claimed selector gain has to clear. On this scale 28% of block-level plug-in effects are *smaller than the noise of re-running the same configuration with a different seed*, and among the blocks in which the plug-in wins, 27% are in that regime — a different seed could have reversed them. The median effect is 2.13× the noise scale, so the typical effect is real but not comfortably so, and single-seed per-block claims of the kind that fill plug-in tables are not identifiable. This is a statement about *blocks*, not about the aggregate: the pooled tests of §4 average over 128 blocks and are correspondingly better powered.

Against compute. Plug-ins are advertised as free, and in parameter count they are: all three add zero trainable parameters and shift peak memory by less than 0.3 MiB. Wall clock is a different matter. **fredf** costs 1.03× the training time of **none** for a median +0.45%; **revin** costs 1.05× for +0.03%; **san_lite** costs 1.73× — a 73% training-time surcharge — to be 3.01% *worse*. The arm with by far the largest compute overhead is the only one that is significantly worse than doing nothing on both metrics. Efficiency framing does not rescue any of the three: the best of them buys 0.13 points of median MSE per 1% of extra training time.

7. Where conditionality lives: below the block, and the one legal axis

Section 4 established conditionality at the level of the block. Conditionality also exists *inside* a block, and examining it answers a question that the rest of the paper cannot: is there any selection axis that deployment does not forbid?

Our evaluation stores per-timestep test MSE aggregated into four quarters of the forecast horizon (**seg**₁ near, **seg**₄ far); the four quarter errors average exactly to the block MSE, so they form an exact decomposition. Two features of this audit matter. First, it covers all 128 blocks and all four backbones, including TimesNet, which the per-window analysis of §8–§9 cannot reach (§13). Second, and more important, *the lead-time index is known at decision time*. A policy that applies one arm to the first half of the forecast and another to the second half needs no feedback and incurs no delay. It is the one axis in this paper on which selection is causally legal.

Table 7

Arm identity is unstable along the forecast horizon, yet the legal axis it opens is empty.

(a) Does one arm win across all four lead-time quarters?				
Stratum	Blocks	Constant argmin	Mean # distinct	Whole-horizon winner suboptimal somewhere
3-arm pool	84	65%	1.43	35%
4-arm pool	44	39%	1.73	61%
DLinear	32	38%	1.72	—
PatchTST	32	62%	1.44	—
TimesNet	32	62%	1.44	—
iTransformer	32	62%	1.53	—
(b) Median relative gain (%) by lead-time quarter				
Arm	Q1 (near)	Q2	Q3	Q4 (far)
revin	+0.14	+0.12	-0.02	-0.03
san_lite	-5.03	-3.49	-2.82	-1.83
fredf	-0.36	+0.32	+0.93	+0.48
(c) Headroom of the one causally legal selection axis				
Reference	Mean (%)	Median (%)	Max (%)	Blocks with gain
none	+3.82	+1.27	—	—
best fixed arm	+0.24	+0.00	+3.29	56/128 (12% > 0.5 pp)

seg; errors partition the *forecast horizon* into quarters (per-timestep MSE averaged within each quarter), not the test period; the four quarter errors average exactly to the block MSE, which is what makes panel (c) a valid decomposition. Stratifying by arm-pool size is necessary: a 4-arm block has more chances to switch than a 3-arm block, so the two strata must not be pooled. This audit covers all 128 blocks and all four backbones, including TimesNet, which the per-window analysis cannot reach. Panel (c) is the decisive one: unlike the per-window axis, the lead-time index is known at decision time, so switching arms along it needs no feedback and is causally legal — but even its *oracle* buys almost nothing over a single frozen arm.

The instability is real (Table 7a). Only 56% of blocks have a single arm that is best across all four quarters, and in 44% of blocks the arm chosen on whole-horizon MSE is beaten by another arm somewhere along the horizon. Stratifying by pool size is necessary — a four-arm block has more opportunities to switch than a three-arm block — and the pattern survives it (65% constant in three-arm blocks, 39% in four-arm blocks). The strata here are 84 and 44 of the 128 *blocks*, which is a different denominator from the 83/43/1 *decision groups* of §3: this audit reads block-level segment errors and therefore covers TimesNet and all 128 blocks, whereas a decision group additionally requires aligned per-window traces and there are 127 of those. The profiles are also interpretable (Table 7b): **fredf** is mildly harmful at short lead times (median -0.36% in Q1) and helpful further out ($+0.93\%$ in Q3), consistent with a spectral term that trades local fidelity for global shape, whereas **san_lite**'s damage is worst near the origin (-5.03% in Q1) and decays with lead time.

And yet on the granularity we can measure, the axis is nearly flat (Table 7c). A *quarter-wise oracle* that picks the best arm independently for each of the four quarters beats **none** by $+3.82\%$ on average, which sounds like a target — but almost all of that is the value of choosing an arm at all, not of switching. Measured against the best single frozen arm in the same block, the quarter-wise lead-time oracle is worth $+0.24\%$ on average, exactly 0.00% at the median, and above half a point in only 12% of blocks; its maximum anywhere in the grid is $+3.29\%$.

Two things this does *not* establish, and we were careless about both in an earlier draft. It is not an upper bound on finer lead-time policies: a per-step oracle partitions the horizon more finely, so its loss is no larger and its headroom is at least $+0.24\%$. And it is not a bound on deployable rules of a *coarser* family either, except in the trivial sense that a quarter-wise oracle bounds quarter-wise rules. The honest statement is that the family we can measure — horizon split into four segments, one frozen arm per segment — has almost no headroom over a single frozen arm, and that measuring finer families would require per-step errors on the validation split, which our pipeline does not persist. Given the size of the quarter-wise number we did

Table 8

Lagged agreement of best-arm identity, 127 groups. The reference used in the excess column is the marginal chance level $\sum_k p_k^2 = 0.395$. Per-window re-labeling drives the measured agreement to 0.307, which matches the uniform-label null $\text{mean}(1/K) = 0.306$ rather than the marginal level, as it must: re-labeling destroys the marginal concentration along with the association. Any lag $\ell \geq H$ is causally legal and $\ell = H$ is the smallest legal lag, which is why the H row is the deployable one; the $2H$ row is legal too and is shown for the trend.

Lag ℓ	agreement	excess	groups $z > 2$
1 window	0.864	+0.469	127/127
$H/4$	0.467	+0.072	93/127
$H/2$	0.415	+0.020	72/127
H (smallest legal)	0.389	−0.006	37/127
$2H$	0.397	+0.002	41/127

not add them; persisting them is a one-line change and we recommend it to anyone who wants to settle this axis rather than infer it.

The two axes therefore fail for opposite reasons on this grid: the per-window axis has large headroom that expires before it can be acted on (§8–§8.2), while the quarter-wise lead-time axis can be acted on freely but has little headroom to act on.

8. Why the evaluated per-window selectors did not work

8.1. The per-window oracle is label-invariant

Every gating paper motivates itself with a per-window oracle: if for each evaluation window t we could pick the arm k with the smallest error $e[k, t]$, we would gain 10%. Our matrix reproduces this: per-window oracle vs. `none` is +10.15% over 127 groups (and +9.74% vs. `none` on the 96 gate-eligible groups, where the deployable `best_fixed_val` gains only +2.12%).

This statistic cannot support the inference that is routinely drawn from it.

Proposition 1 (Re-labeling invariance). *Let $e[\cdot, t] \in \mathbb{R}^K$ be the arm errors in window t and let σ_t be a permutation of $\{1..K\}$ drawn independently for each t . Then $\frac{1}{T} \sum_t \min_k e[\sigma_t(k), t] = \frac{1}{T} \sum_t \min_k e[k, t]$.*

PROOF. $\min_k e[\sigma_t(k), t]$ depends only on the multiset $\{e[1, t], \dots, e[K, t]\}$, which σ_t leaves unchanged.

The oracle therefore measures only the *dispersion* of arm errors within windows, not any association between window and best arm. Independently permuting arm labels per window destroys every learnable pattern and leaves the headroom bit-identical — we verified this numerically in all 127 groups (127/127 invariant). Any non-degenerate arm pool has a large per-window oracle; “oracle $\gg$ best fixed” is a property of variance, not of learnability.

The quantity that does govern learnability is the lagged predictability of the best-arm identity $a_t = \arg \min_k e[k, t]$. We measure the agreement $\Pr[a_t = a_{t-\ell}]$ against the chance level $\sum_k p_k^2$, where p_k is the marginal win frequency of arm k .

Table 8 is the core of the paper. At lag 1 the identity is enormously predictable (+0.469 over chance, every group flagged). At the smallest lag a deployed system may use it is *at chance* (−0.006, and no larger at $2H$). The signal exists; it expires before it can be acted on. Four of the 127 groups — Exchange at $H=720$ on all four backbones — contribute only 78 overlapping pairs each at the legal lag rather than several hundred, and we report in §13 what happens when they are dropped.

The per-group z column is a dispersion diagnostic, not an inference. The $z > 2$ counts use the i.i.d. binomial reference $z = (\widehat{\text{Pr}} - \text{chance}) / \sqrt{\text{chance}(1 - \text{chance}) / (n - \ell)}$, which is anti-conservative here: windows overlap at stride 1, so the $n - \ell$ agreement indicators are far from independent and $|z|$ is inflated by roughly the number of windows per independent block. We keep the column because its *shape* is the point. At the legal lag the counts are exactly symmetric — 37/127 groups with $z > 2$ and 37/127 with $z < -2$, and 57/127 with positive excess — which is the signature of an over-dispersed null rather than of a weak but

Table 9

Excess-over-chance agreement of best-arm identity as a function of absolute lag, pooled over the 127 decision groups.

lag	1	4	16	32	64	128	512
excess	.469	.321	.163	.092	.041	−.007	.005

Table 10

The decay is not an averaging artifact: three independent stratifications (backbone, dataset, horizon) give 20 usable strata and no exception.

Criterion	Result
strata with half-life $< 25\%H$	20/20
strata with excess > 0.05 at lag H	0/20
most signal-like stratum	Exchange, $+0.032$
longest relative correlation	$H=24$, 19.2%
range of median half-life	3.0 – 15.3 windows

real signal. Dividing each z by the crude block factor $\sqrt{H}$ leaves 0/127 groups significant in either direction. By contrast, at lag 1 the 127/127 count is one-sided and survives any such deflation. No claim in this paper rests on these z values; the decisive quantities are the mean excess (-0.006) and the half-life of §8.2.

Why the smallest legal lag is H . Window t predicts the interval $[t + L, t + L + H)$. Its realized error is observable only after that interval closes, i.e. H windows later, so every lag $\ell \geq H$ is legal and $\ell = H$ is the tightest of them. A selector that decides for window t using any feedback from window $t' > t - H$ is using information that does not exist at decision time. This is an arithmetic property of the feedback model rather than a modeling preference, and the model is the right one when the target is a block of H future values scored after the fact. It is the wrong one for a task where partial or proxy feedback arrives sooner — an early-lead-time error, a cheap surrogate, an exogenous regime label — and §11 treats that as the most promising opening we can see. Within our model it is also the reason the zero-delay ablations common in this literature look so much better than anything deployable.

8.2. How long does the signal live?

We summarize the decay of the excess agreement by its *half-life*: the absolute lag (in windows) at which excess-over-chance falls to half of its lag-1 value.

The median half-life is **9.0** windows (mean 11.3). Because H varies across groups we also report the ratio directly: half-life/ H has mean **6.1%** and median **5.1%**. Across 127 groups, 86% have a half-life below $H/10$. The modal arm already wins 48.8% of windows (normalized win-distribution entropy 0.88), so a majority-vote constant policy captures most of the achievable identity mass.

Is this an average artifact? A reviewer will ask whether some backbone, dataset, or short horizon does carry usable signal that the average hides. We recompute the audit under three independent stratifications (backbone, dataset, horizon), 20 usable strata, none under-powered.

Horizon scaling is the wrong way round. The absolute half-life does grow with H (Spearman $+0.74$ over the eight horizon-level medians, $+0.49$ over all 127 groups; 4.6 windows at $H=24$, 15.3 at $H=720$) but far more slowly: H grows $30\times$ while the half-life grows $3.3\times$. Hence half-life/ H *decreases* with H (Spearman -0.88 over the horizon-level medians, -0.73 over all groups): 0.192 at $H=24$ down to 0.021 at $H=720$. Long-horizon forecasting — precisely where plug-ins are marketed — is where the gap between the usable correlation length and the legal decision lag is widest, and so where window-level plug-in selection is least likely to pay. Figures 1 and 2 plot the two views.

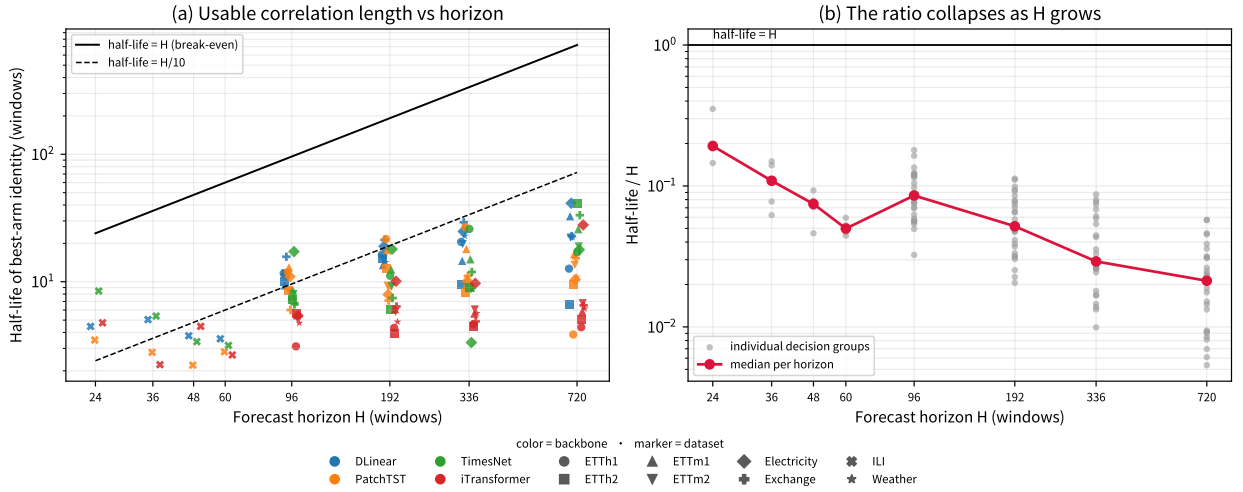


Figure 1: The usable correlation length stays below the decision lag in every group we measured. (a) Half-life of best-arm identity vs. forecast horizon, one marker per decision group (127; color = backbone, marker = dataset), log-log. The solid line is the break-even condition $\text{half-life} = H$; no group reaches it (the maximum is $0.35H$, TimesNet/ILI/ $H=24$), and most sit below the $H/10$ dashed line ($109/127 = 86\%$; the remaining 18 groups fall between $0.1H$ and $0.35H$). (b) The ratio half-life/ H per horizon: gray dots are groups, red is the median, falling from 0.19 at $H=24$ to 0.021 at $H=720$. The decline is monotone *within* each of the two horizon families ($24/36/48/60$ for ILI, $96/192/336/720$ for the rest); the small step up at $H=96$ is the change of family, not a reversal.

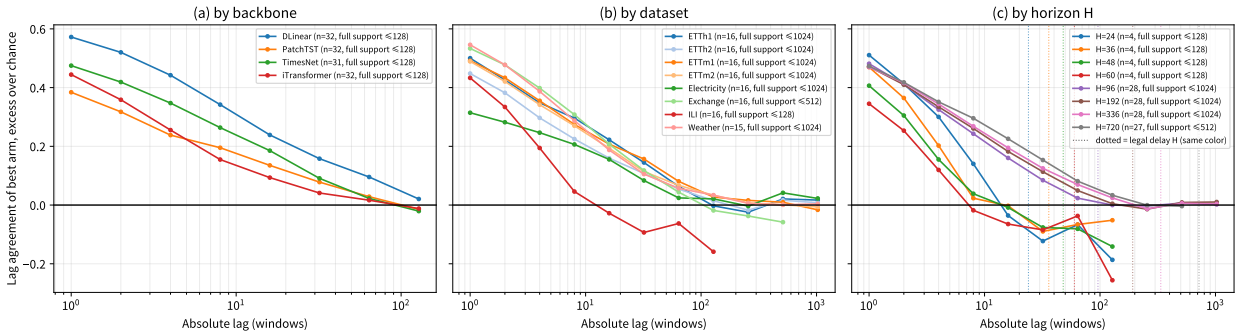


Figure 2: The same decay under three independent stratifications. Excess-over-chance agreement of best-arm identity vs. absolute lag (windows), by backbone (a), dataset (b) and horizon (c). Only lags with full support across the groups in a stratum are drawn. Every curve reaches chance by lag $\approx 10^2$; in (c) the dotted vertical lines mark each stratum's smallest legal decision lag H (same color), and every curve has already flattened at or before its own line. None of the 20 strata is an exception.

8.3. The headroom is real, just late

Proposition 1 is constructive; a skeptic may answer that real labels are not random permutations. We therefore give a purely empirical second leg. For 48 cells we retrain with only the training seed changed ($2021 \rightarrow 2022$) and ask how much of the per-window best-arm identity survives.

The key statistic is the *cross-seed oracle*: choose arms with seed A's per-window arg min, settle the bill on seed B's errors. It still uses future information, but only the part of the arm identity that is reproducible, so it upper-bounds any selector that relies on reproducible structure — including a static gate given perfect features.

The last three rows of Table 11 are a necessary caveat: if a backbone converges to the same solution under both seeds, reproducibility is trivial and the audit has no discriminative power for it (near-convex models such as DLinear are the worst case). With same-arm error correlation 0.989 and block-level MSE

Table 11

A seed-paired replication shows the per-window structure is real and largely reproducible, so the failure of the evaluated selectors is not explained by “there is nothing there to learn”. The replication covers 48 of the 127 decision groups (37.8%) on three backbones — TimesNet is absent — and the reproducible share of headroom is heterogeneous across them: DLinear 99.1%, PatchTST 87.2%, iTransformer 48.7%.

Quantity	Value
paired cells / seeds	48 / {2021, 2022}
same-seed oracle vs. best fixed	+7.26%
cross-seed oracle (reproducible bound)	+5.74%
reproducible share of headroom	79%
cross-seed identity agreement	0.788 (chance 0.374)
pairs with $z > 2$	48/48
pairs where cross-seed oracle still wins	46/48
same-arm per-window error correlation	0.989
block MSE fluctuation across seeds	1.33%
best fixed arm unchanged across seeds	88%

Table 12

Window-level gating. Gains in % relative to the stated reference; bf = best_fixed_val. “in-pool” trains and applies within each group; “LODO” holds out a whole dataset; “learnable” restricts to groups with ≥ 1000 validation windows.

Protocol	head	N	vs none	vs bf	oracle
in-pool	reg	96	+1.50	−0.67	+9.74
in-pool	clf	96	+1.49	−0.65	+9.74
learnable	reg	72	+2.46	−0.66	+9.10
LODO	reg	76	+1.23	−1.56	+10.46

moving by 1.33%, the model does change, yet *which window wants which plug-in* stays stable. The backbone breakdown in the caption is the honest version of this: the 79% pooled figure is carried by DLinear and PatchTST, and for iTransformer under half the headroom survives a seed change, so for that backbone part of the oracle number *is* training noise. So the finding is not “there is nothing to learn” but the more specific “on most of the grid the thing to learn is real, and it is unavailable in time”.

9. Three selector families, one verdict

We now test the decision problem directly. All numbers are relative to the deployable best_fixed_val baseline on identical decision groups.

9.1. Static complexity-feature gating

Per window we compute 24 cheap statistics in five groups — non-stationarity (**w_ns**), trend and autocorrelation (**w_sd**), cross-channel correlation (**w_ch**), spectral concentration (**w_fr**) and predictability/entropy (**w_pc**) — all $O(n \log n)$ from the raw look-back window, with no access to the target. A gradient-boosted tree is trained on validation windows (either regressing per-arm gain and thresholding, or classifying the arm directly) and applied to test windows.

Gating loses to a frozen arm by 0.67 points (positive in 22% of groups, Wilcoxon signed-rank across groups $p=3.1 \times 10^{-6}$), captures a *negative* fraction of the oracle headroom, and its per-window accuracy (0.416) is below the majority-class rate (0.486). Restricting to data-rich groups does not help (−0.66); requiring cross-dataset transfer makes it worse (−1.56). Stratifying by the number of validation windows (< 200 , $200\text{--}1000$, $1000\text{--}5000$, ≥ 5000) gives −0.46, −1.09, −0.60, −0.73: the failure is not a sample-size artifact. The LODO row covers 76 of the 96 gating groups: leaving out one dataset requires at least three datasets sharing a (backbone, horizon, arm-pool) shape, which excludes ILI entirely (each ILI horizon exists

Table 13

Within-split chronological diagnostic: gate fitted and evaluated inside one split (70/30 by time), removing label optimism and the val→test source difference but not drift and not target overlap at the boundary. bf_{split} is the fixed arm chosen on the fitting segment. Groups with < 200 fitting or < 100 evaluation windows are skipped. The two rows are different group sets — val-internal covers three backbones and 81 groups, test-internal four backbones and 111 — so -1.18 and -0.53 are not a paired contrast.

fit/eval split	N	vs none	vs bf_{split}	win	oracle
val internal	81	+2.01	-1.18	35%	+11.35
test internal	111	+2.33	-0.53	38%	+10.37

on one dataset only) and excludes the four-arm PatchTST/iTransformer shapes at $H=96$ and $H=720$, which span two datasets. The exclusions are structural, not outcome dependent.

Falsification controls. Two controls separate “window-level information” from “switching per se”: a random switcher matched to the gate’s switching frequency and arm marginals (-1.04 vs bf), and the same model trained on column-shuffled features, i.e. a same-capacity noise-fitting baseline (-0.54). The gate’s net advantage is $+0.36$ over random (66% of groups) and -0.22 over shuffled (26%). The honest reading: after the overfitting baseline is subtracted, the contribution of window-level features is not distinguishable from zero, and we do not claim it. Leave-one-group-out feature ablation is consistent — removing any of the five feature groups changes the gain by at most 0.13 points, and removing the entropy group even *improves* it.

Within-split chronological diagnostic. The gate above is trained on validation windows and applied to test windows, so two confounds remain: validation errors are mildly optimistic (that split also drives early stopping) and there is a val→test source difference. We remove both by re-running the identical gate *inside a single split*: the first 70% of windows chronologically to fit, the last 30% to evaluate, with the fixed-arm baseline bf_{split} also chosen on the fitting segment, so neither side sees the evaluation segment. With **source=val** the optimism is present on both sides and cancels; with **source=test** the labels have *never* been touched by early stopping or checkpoint selection, so the label bias is removed and all backbones are covered. Both are diagnostic upper bounds, not deployable policies.

Two things this diagnostic does not remove, and the earlier draft’s phrase “zero drift” was wrong about both. It removes the val→test *source* difference, not distribution drift: on the 84 groups where we can recover the fitting-segment error of **none**, its MSE differs between the two segments by a median of 25.5% in absolute terms (mean 40.0%, maximum +155.4%, and the sign runs both ways), with 61 of 84 groups above 10%. And it does not purge the boundary: at stride 1 the last fitting window and the first evaluation windows share forecast targets, so without a purge of at least H origins a median of 22.2% of evaluation windows overlap the fitting segment’s targets (9 of 84 groups overlap completely, and requiring an $L + H$ gap raises the median to 25.0%). Both effects make the diagnostic *easier* for the gate than deployment would be, which is why we read it as an upper bound; a purged version is the natural refinement and we did not run it.

The verdict is unchanged (Table 13). With the label optimism and the source difference removed — and with a boundary that if anything favours the gate — it still loses to a frozen arm chosen from the same information (-0.53 , positive in 38% of groups, $p=0.19$), and its per-window accuracy (0.411) is still below the majority-arm rate (0.486), beating it in only 13% of groups. Its share of the +10.4% oracle headroom is small and, more importantly, unstably estimated: as a ratio of aggregate means it is $+2.33/+10.37 = 22\%$, whereas the per-group ratio has median 4.6% and mean 3.3% with an interquartile range of -9% to $+35\%$ and a minimum of -579% , because groups whose oracle headroom is nearly zero put a vanishing denominator under a noisy numerator. We report all three rather than pick one, and we do not rest any conclusion on them: the load-bearing fact is the comparison against the frozen arm, which needs no ratio. This rules out “the validation split was used for early stopping” and “the val→test source difference” as explanations: what is missing is window-level signal, not clean labels. One further detail sharpens the main result: in the test-internal setting the gate is on par with the *validation*-chosen arm ($+0.08$) while losing to the split-chosen arm (-0.53), and the split-chosen arm coincides with the evaluation-segment argmin in 70% of groups versus 47%

Table 14

Online selection with legal H -window delay (123 groups). Pool sizes are nominal; realized arm counts per group are given in §3. “delay cost” is the no-delay advantage.

Pool (nominal)	vs none	vs bf	no-delay	delay cost
≤ 4 arms	+1.19	−0.87	+2.47	3.34 pp
≤ 11 arms	+1.31	−0.82	+2.84	3.66 pp

Table 15

Delay $\times$ feedback-window sweep, gains vs. bf (%). “fb” is the sliding feedback-window length. Lags $\ell \geq 1H$ are causally legal; shorter ones are shown to locate the break-even point.

delay	fb= 50	200	1000	3000	legal?
0	+4.59	+2.19	+0.96	+0.88	cheat
0.25 H	+0.83	+0.07	+0.22	+0.43	cheat
0.5 H	−0.49	−0.64	+0.01	+0.32	cheat
1H	−1.36	−0.77	−0.13	+0.12	yes
2 H	−0.92	−0.61	−0.01	+0.22	yes

in the val-internal setting. The small gains a window gate appears to make over `best_fixed_val` elsewhere are arm-identity mismatch between splits, not per-window adaptation.

9.2. Delayed online error feedback

Static features must transfer across a distribution shift; deployment has a stronger signal available, namely the *realized* per-arm error on past windows. We run follow-the-leader over a sliding feedback window of 200, with the delay set to exactly H windows, and additionally a no-delay cheating variant. This is the delayed-feedback regime of [13], with the distinguishing feature that the delay is not an implementation nuisance but is fixed by the forecast horizon itself.

The legal configuration loses (−0.87, positive in 20% of groups) while the illegal one wins by +2.47: a 3.34-point delay cost. This is the same conclusion as Table 8, reached with a different estimator and no learned model at all. The count is 123 of the 127 decision groups; the four excluded ones are Exchange at $H=720$ on all four backbones, where a 720-window delay leaves only 78 post-delay decisions out of 798 test windows. We exclude them because 78 decisions is too few for a stable estimate, not because no legal decision exists — an earlier draft said the latter, which was wrong. Including them moves the legal mean from −0.87 to −0.85 and changes no sign. These four groups do enter the other audits, where their support is also thin: at the legal lag in Table 8 each contributes 78 overlapping pairs, and dropping them moves the mean excess from −0.006 to −0.012 and the $z > 2$ count from 37 to 35.

Delay sweep. Sweeping the delay in multiples of H (all cells evaluated on one common test segment, skipping the warm-up of the longest delay — omitting this dilutes long-delay rows toward zero and produces the spurious conclusion that $2H$ beats $1H$) gives a clean break-even point (Table 15).

Averaged over feedback windows on the 103 groups with common support, the break-even delay is **0.25 $\times H$** : at $1 \times H$ the policy is −0.38%, at zero delay +1.89%. Note also the interaction: short feedback windows are best when the delay is small and worst when it is legal, because adaptation speed is only useful if the information is fresh.

9.3. Horizon-aware checkpoint selection

The third family is zero-compute: instead of selecting the checkpoint by overall validation MSE, we select it by a horizon-aware read of the same training logs (worst forecast segment, long-segment MSE, slope-penalized, rank aggregation, last epoch). Over all 844 cells, none beats the validation-MSE baseline (gains −0.002 to −0.074; taking the last epoch is −2.03, $p=6.8 \times 10^{-13}$), while the post-hoc oracle checkpoint would give +0.92. Even re-reading existing logs yields nothing.

Table 16

The same three effects at three levels of aggregation. “pooled” is the mean over decision groups, as in Tables 12–14; “ $\pm$ datasets” is a 95% percentile interval from a cluster bootstrap that resamples the eight datasets; LODO is the range of the pooled mean over the eight leave-one-dataset-out subsets; p_{ds} is an exact Wilcoxon signed-rank test on the eight dataset means, whose two-sided floor at $n=8$ is 0.0078. Generated from `artifacts/cluster/`.

Effect	pooled	$\pm$ datasets (95%)	sign	LODO range	p_{ds}
per-window oracle vs none	+10.15	[+6.38, +13.91]	8/8	[+9.21, +11.45]	0.0078
gate (in-pool) vs bf	−0.67	[−0.94, −0.37]	1/8	[−0.78, −0.59]	0.0156
online (legal H) vs bf	−0.87	[−1.75, +0.07]	1/8	[−1.24, −0.59]	0.0781

10. What the eight datasets support

Every number so far is aggregated over decision groups, and a group is a (backbone, dataset, horizon) triple. That is the right unit for the question “did this selector beat a frozen arm in this configuration”, and the wrong unit for the question a reader actually cares about, which is whether a selector works on datasets it has not seen. The 123 online groups live on eight datasets, so sixteen of them share a dataset with fifteen others; treating them as independent evidence is the standard mistake in this literature and we made it in an earlier draft, where a group-level Wilcoxon test on the online effect returned $p = 1.2 \times 10^{-5}$. That p -value assumes 123 independent units and is not defensible.

We therefore re-report the three load-bearing effects with the dataset as the resampling unit: a nonparametric cluster bootstrap over datasets (20,000 resamples), a leave-one-dataset-out sweep, and an exact paired test on the eight dataset means. Table 16 gives all three.

Three things change and one does not.

The oracle headroom survives. All eight datasets have a positive mean, the cluster interval is far from zero, and the leave-one-dataset-out range [+9.21, +11.45] is narrow — dropping Electricity, the least favourable dataset, moves it up rather than down. The claim “there is large per-window headroom on this grid” is a cross-dataset claim and it holds. Combined with Proposition 1, that is exactly the uncomfortable pairing the paper is about: a robust number that is not evidence of learnability.

The gating loss survives. Seven of eight datasets are negative, the cluster interval excludes zero, and no leave-one-out subset changes the sign. The best dataset, Weather, is +0.15, which is inside the seed noise band of §6.

The online loss weakens to a directional result. Seven of eight datasets are negative and every leave-one-out subset stays negative, but the dataset-cluster interval is [−1.75, +0.07] and the exact dataset-level test gives $p = 0.078$, which at $n = 8$ is the second-smallest attainable value. The pooled point estimate −0.87 is unchanged; what changes is what we may say about it. We therefore state it as: *on seven of eight datasets the legal online selector lost to the frozen arm, and we cannot exclude a small positive effect on a new dataset.* ETTh1 is the single positive dataset (+1.62) and also the most influential one — dropping it moves the pooled mean to −1.24.

Nothing reverses. No effect changes sign under any leave-one-dataset-out subset, and the dataset-level Friedman test over the four arms ($\chi^2 = 7.80$, $p = 0.050$, $n = 8$) is at the edge of significance with a Nemenyi critical difference of 1.66 that is wider than the entire range of average ranks. So the *ranking* of the four arms is not established across datasets either, and we do not claim one.

Two structural limits on this analysis. First, eight clusters is few. The bootstrap intervals are wide because they should be, and an exact paired test on eight datasets cannot report anything below $p = 0.0078$; we quote that floor next to every dataset-level p -value rather than let a reader read 0.0156 as strong. Second, a two-level (dataset, seed) resampling is not available to us: the three-seed replicates cover only four datasets and the per-window traces exist for seed 2021 alone, so the seed floor of §6 and the dataset clustering here cannot be combined into one interval. We report them side by side and note that both point the same way for the two selector effects, whose magnitudes (0.67 and 0.87 points) sit inside the 0.30–2.41% interquartile band of within-configuration dispersion.

Table 17

Decomposition of the online policy’s gain over **none**. The time-varying component is negative; the whole effect is the correction of a constant mis-selection.

Component	Value
bf_test – bf_val	+1.36%
(groups where val picks the wrong arm)	34%
online – bf_test	–2.32%
(groups where this is positive)	5%

Absolute error, not only relative gain. One asymmetry is easy to hide behind percentages. “Positive in 22% of groups” does not mean “worse in 78%”: some groups tie exactly, because the selector happens to reproduce the frozen arm. Reading the absolute MSE instead, the in-pool gate is strictly worse than **bf** in 68 of 96 groups (70.8%), by a median of +0.0046 MSE and up to +0.096 (iTransformer/ILI/ $H=24$, 2.294 $\rightarrow$ 2.390), and the legal online selector is strictly worse in 79 of 123 groups (64.2%), by a median of +0.0043 and up to +0.079 (DLinear/ETTh2/ $H=720$, 0.522 $\rightarrow$ 0.601, a 25.7% relative regression). The classifier-headed gate and the LODO gate are worse in 70.8% and 77.6% of groups. Absolute MSE is not comparable across datasets — ILI is an order of magnitude above ETT — so we use it only for the direction and the per-group worst case, which is the number a practitioner deciding whether to deploy a gate should see.

11. What is actually recoverable

If online feedback loses to **best_fixed_val**, what did its +1.19% over **none** consist of? Decomposing against the post-hoc constant arm **best_fixed_test** settles it (Table 17).

The time-varying component is negative. What is left is that validation-based arm choice is wrong in a third of groups, and delayed feedback partially corrects that *constant* choice. This is a modest, deployable claim: *use online feedback to fix plug-in mis-selection, not to adapt per window*. A paper claiming window-level adaptation should publish the **best_fixed_test** column, because without it the reader cannot tell the two apart.

Two openings we did not close. Both follow from where our evidence stops rather than from where it points. *Different feedback.* The whole audit assumes the selector’s signal is the realized per-window error, which arrives one horizon late. A signal that arrives sooner — the error at early lead times within an in-flight forecast, a cheap surrogate for it, or an exogenous regime indicator that needs no feedback at all — is not subject to the delay and is not tested here. The correlation-length measurement says the *lagged realized error* is uninformative at the legal lag; it says nothing about a proxy available at decision time. *Coarser granularity.* Between the per-window axis, which has headroom but no usable memory, and the per-run axis, which is what **best_fixed_val** already does, there is a range of segment lengths we did not sweep: switch once per H windows, once per month, once per detected regime. The half-life of 9 windows is short relative to H , but a policy that commits for $k \gg$ half-life windows is not asking the identity to persist — it is asking the *marginal* best arm to drift slowly, which our data does not rule out. Both would reuse the released per-window arrays and neither needs new training.

12. A checklist for adaptive-selection claims

The audit generalizes beyond plug-ins to any “learn which model/module to use when” claim. We propose seven tests, all cheap once per-window errors are persisted:

1. **Re-labeling test.** Recompute your oracle after permuting arm labels independently per window. If it does not move (Prop. 1), it is not evidence of learnability and must not be cited as headroom.
2. **Correlation-length test.** Report the half-life of best-arm identity in units of the decision lag. If half-life/lag $\ll 1$ (here: mean 0.061, median 0.051, < 0.25 in 20/20 strata), a selector that depends

only on lagged realized feedback has very little to work with, and none of the three families we evaluated overcame it. The test is diagnostic, not a proof of impossibility: a selector using information other than lagged feedback is not covered by it.

3. **Deployable-baseline test.** Compare against the arm frozen from validation, and publish the post-hoc constant arm too, so that “adaptation” is separated from “fixing a bad constant choice” (§11).
4. **Shuffled-feature test.** Train the identical model on column-shuffled features. The difference, not the raw gain, is your evidence (here: -0.22 points).
5. **Within-split test.** Before blaming the val→test source difference or noisy labels, fit and evaluate the selector inside one split, chronologically, with the fixed-arm baseline chosen on the same fitting segment, and purge at least H forecast origins at the boundary so the two segments do not share targets. A selector that loses in this easier setting (Table 13) is unlikely to win in deployment, and this costs no additional training. Report the drift that remains: the setting removes the source difference, not the shift (here median 25.5% in the baseline arm’s error level).
6. **Absolute-regression test.** Report how often the selector’s *absolute* error is worse than the frozen arm’s, not only how often its relative gain is positive. The two are not complements: our in-pool gate is positive in 22% of groups but strictly worse in 71% (median $+0.0046$ MSE, maximum $+0.096$), and the legal online selector is positive in 20% but strictly worse in 64% (median $+0.0043$, maximum $+0.079$). The remainder are ties, and a reader who sees only the positive share will over-estimate the downside protection.
7. **Dataset-unit test.** If the claim is that a selector works “in general”, resample datasets rather than configurations, and report a leave-one-dataset-out sweep. With eight datasets an exact paired test cannot go below $p = 0.0078$, so say so instead of quoting a group-level p of 10^{-6} that assumes 123 independent units (§10).

Tests 1–2 say whether the target is reachable, test 5 whether the pipeline is to blame, tests 3–4 and 6 whether a reported win is a win, and test 7 whether it is a win anywhere but the datasets it was tuned on.

For the reporting side, §4–§6 add four more that cost nothing but a few lines of analysis code: aggregate with pairwise dropping and publish the per-arm block counts; report every verdict on at least two error metrics, since one of our three arms changes verdict between MSE and MAE; report the median, the mean and the tail fractions together, because the first two disagree in sign for one of our arms and the tail is where the deployment risk lives; and put the effect next to the run-to-run noise of a training seed and next to the wall-clock time it costs, because 28% of our block-level effects are inside seed noise and the arm with the largest compute surcharge is the one that hurts most. Finally, report any plug-in both at its published default and per-cell tuned, with the gap stated (§5).

13. Threats to validity

A provenance bug we found in ourselves. An earlier version of this study trained the window gate on validation error traces whose order had been permuted by a shuffling data loader (the library disables shuffling for `test` but not for `val`). Features of window i were paired with the error of a random window j . Every gating/control/ablation number was therefore invalid, while all test-side audits were unaffected (the test loader never shuffles) and all per-arm *means* were unaffected (a mean is permutation-invariant, so `best_fixed_val` survived). We re-ran 328 cells, added a machine-checked provenance marker asserting deterministic loader order, and made the analysis code refuse unmarked arrays rather than silently reuse them. We report this because the corrected gating number (-0.67) is smaller in magnitude than the buggy one (-1.76): the bug was making our negative result look stronger than it is.

Coverage asymmetry. The test-side audits (Tables 8, 14) use 127 decision groups; only 96 have a certified deterministic validation window order and can therefore train a gate (Table 12). Table 18 makes the gap explicit rather than asserting it is harmless. All 31 missing groups are `TimesNet`, for a bookkeeping reason: those validation traces predate the provenance marker above. The question is whether they differ on the quantities that decide learnability. They do not: half-life over the legal delay is 0.053 versus 0.049 for covered groups, and excess above chance at delay H is -0.002 versus -0.008 — the same distribution, both far from

Table 18

Explicit coverage gap between the test-side audits and the gating protocols, with the audit statistics that decide learnability. $t_{1/2}/H$ is the median arm-identity half-life divided by the legal feedback delay; $\text{exc}@H$ is the mean excess above chance at delay H .

backbone	audit	gate	oracle	$t_{1/2}/H$	$\text{exc}@H$
DLinear	32	32	+14.73	0.076	-0.010
PatchTST	32	32	+4.66	0.047	-0.010
iTransformer	32	32	+9.83	0.030	-0.005
TimesNet	31	0	+11.40	0.053	-0.002
covered	96	96	+9.74	0.049	-0.008
missing	31	0	+11.40	0.053	-0.002

the threshold at which per-window selection could pay. The missing groups have a *larger* oracle headroom (+11.4 vs +9.7), which is precisely the direction that would make gating look good if headroom were the operative variable; it is not. Filling the gap would add samples, not reverse the sign. Two independent audits also cover all four backbones including TimesNet and reach the same verdict: the test-internal split diagnostic (Table 13, -0.53 there) and the lead-time audit of §7, which uses all 128 blocks.

Datasets as the scarce unit. The grid has 844 phase-one runs and 123 online decision groups but only eight datasets, and a cross-dataset claim has eight units, not 123. §10 reports every headline effect with the dataset as the resampling unit for that reason; the practical consequence is that the online result is directional rather than significant at the dataset level, and that we cannot rank the four arms across datasets at all. Adding datasets, not seeds or horizons, is the cheapest way to strengthen or overturn anything in this paper.

Other caveats. (i) A small number of windows produce NaN features when a channel is constant inside the window; the tree model handles missing values natively and we do not impute. (ii) `san_lite` is a training-free surrogate, not the original method, and the compute surcharge we report in §6 is that of the surrogate. (iii) The $\alpha=0.5$ reference comes from phase 1/2, not the sweep; we aligned seeds and checked the reproducibility of duplicate cells. (iv) Missing cells concentrate in the expensive corner, so every cross-backbone average in this paper is computed on a single common block set. (v) The lead-time audit of §7 resolves the horizon into quarters, not individual steps. A per-step oracle is a refinement of the quarter-wise one, so its headroom is at least the +0.24% we measure and could be larger; we have no measurement of how much larger, because per-step validation errors were not persisted. The quarter-wise number is therefore a statement about quarter-wise policies, not a ceiling on all lead-time policies, and the corresponding claim in §7 is stated that way. (vi) `revin` was deliberately pruned outside DLinear (§3), so its metric-level verdict in Table 3 rests on 44 blocks that are predominantly one backbone, and we do not generalize it; the 12-cell verification subset supports the no-op prediction but is too small to support a metric-level claim. (vii) All conclusions are for the four backbones and eight datasets studied at look-back 96 (36 for ILI); zero-shot foundation models are out of scope (they require bf16, unavailable on the target hardware, and would introduce an incomparable baseline).

14. Conclusions

Plug-in gains for time series forecasting are strongly conditional — the same frequency-domain loss moves a linear backbone by +4.6% and a patch transformer by -16.6% — and the way the community aggregates, measures, summarizes and tunes them inflates the reported effect past the effect itself: block filtering, the metric, the choice of mean or median, and the default hyper-parameter each move the headline by more than the claim. That conditionality invites a selector, and a per-window oracle appears to promise 10% for building one.

We showed that the promise does not mean what it is taken to mean: the oracle is invariant to per-window arm re-labeling, so its size measures the dispersion of arm errors within a window and not any learnable association. The underlying structure is nonetheless real and largely reproducible across training seeds —

on most of the grid; for **iTransformer** under half of it survives a seed change. What makes it unusable here is timing. The identity of the best arm decorrelates with a median half-life of 5.1% of the forecast horizon (mean 6.1%), while the feedback model that a horizon- H forecaster is scored under admits no decision lag shorter than H . Three selector families lose to the arm frozen on validation, the break-even delay is a quarter of a horizon, and a quarter-wise oracle on the one axis that carries no delay — the lead-time index — is worth +0.24%. What remains recoverable in our data is the correction of a constant mis-selection.

We are deliberate about the size of that conclusion. It is an audit of one protocol: three plug-ins on four backbones and eight datasets, a feedback model in which realized error arrives one horizon late, three selector families, and MSE and MAE on the standard splits. Read with datasets as the unit of generalization — which is the unit a claim about “plug-in selection” rests on — the oracle headroom and the gating loss hold across all eight, while the online loss is directional (7/8 datasets negative, 95% CI $[-1.75, +0.07]$). We do not conclude that per-window plug-in selection is impossible, and §11 names the two openings we would look at first: feedback that is not the full realized error — an early-lead-time proxy, a cheap surrogate, an exogenous regime label — and selection at a granularity coarser than the window but finer than the run. What we do conclude is that a per-window oracle number, on its own, is not evidence that any of this works, and that the seven audits of §12 cost less than the gate they replace.

CRediT authorship contribution statement

Mianhan Liu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Chen Chen:** Validation, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large language model to assist with copy-editing, LaTeX formatting and the drafting of analysis and plotting scripts. All experimental design, statistical decisions, numerical results and scientific conclusions were produced and verified by the authors, who take full responsibility for the content of the publication.

Relation to other work by the authors

This study and the companion manuscript [17] share an experimental infrastructure and a research programme but address disjoint questions and report disjoint results: this paper asks whether per-window plug-in selection pays under a legal feedback delay, and the companion asks whether a stored per-sample artifact is in the order its consumer assumes. To let editors and reviewers assess the overlap directly we state it explicitly. *Shared:* the forecasting grid and the trained checkpoints that produced the per-window error arrays, the dataset preparation code, and the loader-order defect described in §13, which is reported identically in both manuscripts. *Exclusive to this paper:* the reporting-lever analysis, the α sweep, the seed-noise and compute accounting, the re-labeling proposition, the correlation-length and half-life audit, the three selector families and their controls, the lead-time audit, the dataset-level re-analysis of §10, and the checklist. *Exclusive to the companion:* the statistical order diagnostics, the negative-control degeneracy result, and the provenance contract, of which this paper uses only the resulting deterministic-order marker. No contribution

is claimed in both manuscripts, and neither manuscript's conclusions depend on the other being accepted. The two releases are versioned and archived independently.

Data availability

All datasets used in this study are public standard benchmarks (ETT, Electricity, Exchange, Weather, ILI) and are not redistributed here. The artifacts required to recompute every number in this paper are released at <https://github.com/oscarliu2019/plugin-selection-audit>: the experiment configurations; per-run result tables for the 844 phase-1 and 423 phase-2 runs; per-block aggregates for the 768 phase-3 runs of the α sweep; the window-level feature table; the per-group outputs of every audit reported here (decay curves and half-lives, oracle re-labeling, delayed-feedback and lead-time sweeps, seed replication, split diagnostics, coverage gap); and the audit, statistics, table and figure generators. The raw per-window error arrays written during training are not redistributed; they are regenerated by re-running the phase-2 configuration with the `--save-window-mse` flag. Each table states the artifact it is regenerated from, and the commands are listed in the repository README. A verification script, `tools/verify_paper_numbers.py`, re-derives every quantitative claim in this manuscript from those artifacts and fails on any mismatch; it prints the claim identifier, the section, the value printed here and the value recomputed.

The repository is also archived permanently in Software Heritage: the revision from which this manuscript was produced is `swh:1:rev:ec90998cd56de249ab48eb381934a93a82d3e12a` (origin <https://github.com/oscarliu2019/plugin-selection-audit>, visit `swh:1:snp:cd0b09f87d11cd06b53af22f6b0439fd8687a009`).

The exact release from which every number in this manuscript was computed is archived, with all artifacts, at <https://doi.org/10.5281/zenodo.22809153> (DOI: 10.5281/zenodo.22809153); the repository above is its development mirror.

References

- [1] T. Kim, J. Kim, Y. Tae, C. Park, J.-H. Choi, J. Choo, Reversible instance normalization for accurate time-series forecasting against distribution shift, in: International Conference on Learning Representations (ICLR), 2022.
- [2] Z. Liu, M. Cheng, Z. Li, Z. Huang, Q. Liu, Y. Xie, E. Chen, Adaptive normalization for non-stationary time series forecasting: A temporal slice perspective, in: Advances in Neural Information Processing Systems (NeurIPS), 2023.
- [3] J. Chen, J. E. Lenssen, A. Feng, W. Hu, M. Fey, L. Tassiulas, J. Leskovec, R. Ying, From similarity to superiority: Channel clustering for time series forecasting, in: Advances in Neural Information Processing Systems (NeurIPS), 2024.
- [4] H. Wang, L. Pan, Z. Chen, D. Yang, S. Zhang, Y. Yang, X. Liu, H. Li, D. Tao, FreDF: Learning to forecast in the frequency domain, in: International Conference on Learning Representations (ICLR), 2025.
- [5] X. Qiu, J. Hu, L. Zhou, X. Wu, J. Du, B. Zhang, C. Guo, A. Zhou, C. S. Jensen, Z. Sheng, B. Yang, TFB: Towards comprehensive and fair benchmarking of time series forecasting methods, Proceedings of the VLDB Endowment 17 (9) (2024) 2363–2377.
- [6] L. Brigato, R. Morand, K. J. Strømmen, M. Panagiotou, M. Schmidt, S. Mougiakakou, There are no champions in supervised long-term time series forecasting, Transactions on Machine Learning Research (2026).
- [7] Y. Wang, H. Wu, J. Dong, Y. Liu, C. Wang, M. Long, J. Wang, Deep time series models: A comprehensive survey and benchmark, arXiv preprint arXiv:2407.13278 (2024).
- [8] A. Zeng, M. Chen, L. Zhang, Q. Xu, Are transformers effective for time series forecasting?, in: AAAI Conference on Artificial Intelligence, 2023.
- [9] J. Demšar, Statistical comparisons of classifiers over multiple data sets, Journal of Machine Learning Research 7 (2006) 1–30.
- [10] N. Cesa-Bianchi, G. Lugosi, Prediction, Learning, and Games, Cambridge University Press, 2006.
- [11] V. Cerqueira, L. Torgo, F. Pinto, C. Soares, Arbitrage of forecasting experts, Machine Learning 108 (6) (2019) 913–944.
- [12] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, A. Bouchachia, A survey on concept drift adaptation, ACM Computing Surveys 46 (4) (2014) 44:1–44:37.
- [13] P. Joulani, A. György, C. Szepesvári, Online learning under delayed feedback, in: International Conference on Machine Learning (ICML), 2013.
- [14] Y. Nie, N. H. Nguyen, P. Sinthong, J. Kalagnanam, A time series is worth 64 words: Long-term forecasting with transformers, in: International Conference on Learning Representations (ICLR), 2023.
- [15] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, M. Long, TimesNet: Temporal 2d-variation modeling for general time series analysis, in: International Conference on Learning Representations (ICLR), 2023.
- [16] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, M. Long, iTransformer: Inverted transformers are effective for time series forecasting, in: International Conference on Learning Representations (ICLR), 2024.

- [17] M. Liu, C. Chen, Silent order corruption in per-sample evaluation artefacts: control degeneration, diagnostic blind spots, and a provenance contract, Companion manuscript; artefacts at <https://github.com/oscarliu2019/order-provenance-audit> (2026).